

SQL PRACTICAL TEST

Schema & Sample Data

All questions use the three tables below. Read the data carefully before writing your queries.

Table 1: doctors

doctor_id	doctor_name	specialty	supervisor_id	salary	join_date
1	Dr. Ahuja	Cardiology	4	180000	2018-03-10
2	Dr. Bose	Cardiology	4	165000	2019-07-22
3	Dr. Chopra	Neurology	5	175000	2017-11-05
4	Dr. Dixit	Cardiology	NULL	200000	2015-06-01
5	Dr. Eapen	Neurology	NULL	210000	2014-09-15
6	Dr. Farooq	Orthopedics	7	155000	2020-02-28
7	Dr. Gupta	Orthopedics	NULL	195000	2016-04-12
8	Dr. Hora	Neurology	5	160000	2021-08-19
9	Dr. Iyer	Orthopedics	7	148000	2022-01-30
10	Dr. Joshi	Cardiology	4	170000	2020-05-14

Table 2: patients

patient_id	patient_name	age	doctor_id	admit_date	discharge_date	ward
101	Ramesh K.	45	1	2024-01-05	2024-01-12	Cardiac
102	Sunita M.	60	2	2024-01-08	2024-01-20	Cardiac
103	Priya N.	35	3	2024-01-15	2024-01-22	Neuro
104	Anil S.	70	1	2024-02-01	2024-02-10	Cardiac
105	Kavita R.	50	5	2024-02-05	2024-02-18	Neuro
106	Manoj T.	55	6	2024-02-12	2024-02-25	Ortho
107	Deepa V.	40	3	2024-03-01	2024-03-08	Neuro
108	Suresh P.	65	7	2024-03-05	2024-03-20	Ortho
109	Anita L.	48	2	2024-03-10	2024-03-22	Cardiac
110	Vikram D.	58	8	2024-03-18	2024-03-30	Neuro
111	Geeta W.	42	9	2024-04-02	2024-04-15	Ortho
112	Rajan H.	52	6	2024-04-08	2024-04-18	Ortho

113	Meena C.	38	10	2024-04-15	2024-04-25	Cardia c
114	Harish B.	63	1	2024-05-01	2024-05-14	Cardia c
115	Lata F.	47	5	2024-05-06	2024-05-20	Neuro

Table 3: treatments (Note: patients 103 and 110 have no treatment records)

treat_id	patient_id	treatment	treat_date	cost
1	101	Angioplasty	2024-01-06	85000
2	101	ECG	2024-01-08	2000
3	102	Echocardiogram	2024-01-09	12000
4	102	Medication	2024-01-12	5000
5	104	Bypass Surgery	2024-02-03	150000
6	104	ECG	2024-02-05	2000
7	105	EEG	2024-02-06	8000
8	105	MRI Brain	2024-02-10	18000
9	106	Knee Replacement	2024-02-13	120000
10	107	CT Scan	2024-03-02	9000
11	108	Hip Replacement	2024-03-07	130000
12	109	Angioplasty	2024-03-12	85000
13	111	Arthroscopy	2024-04-04	45000
14	112	Knee Replacement	2024-04-10	120000
15	113	ECG	2024-04-16	2000
16	113	Medication	2024-04-18	3500
17	114	Bypass Surgery	2024-05-03	150000
18	114	ECG	2024-05-05	2000
19	115	MRI Brain	2024-05-08	18000
20	115	EEG	2024-05-12	8000

Questions

Q1. For each medical specialty, calculate the total number of doctors, their average salary (rounded to 2 decimal places), and total salary. Show only specialties where the average salary is greater than 170,000. Sort by average salary descending.

Expected Output:

specialty	doctor_count	avg_salary	total_salary
Neurology	3	181666.67	545000
Cardiology	4	178750.00	715000

Write your SQL query below:

Q2. List every doctor along with their supervisor's name. Doctors who have no supervisor must still appear in the result with NULL shown in the supervisor column. Order by doctor ID.

Expected Output:

doctor_name	specialty	supervisor_name
Dr. Ahuja	Cardiology	Dr. Dixit
Dr. Bose	Cardiology	Dr. Dixit
Dr. Chopra	Neurology	Dr. Eapen
Dr. Dixit	Cardiology	NULL
Dr. Eapen	Neurology	NULL
Dr. Farooq	Orthopedics	Dr. Gupta
Dr. Gupta	Orthopedics	NULL
Dr. Hora	Neurology	Dr. Eapen
Dr. Iyer	Orthopedics	Dr. Gupta
Dr. Joshi	Cardiology	Dr. Dixit

Write your SQL query below:

Q3. For every doctor, show their name, specialty, total number of distinct patients they treated, and the total cost of all treatments given to their patients. Doctors with no patients must appear with 0 for both counts. Order by doctor ID.

Expected Output:

doctor_name	specialty	patient_count	total_treatment_cost
Dr. Ahuja	Cardiology	3	391000
Dr. Bose	Cardiology	2	102000
Dr. Chopra	Neurology	2	9000
Dr. Dixit	Cardiology	0	0
Dr. Eapen	Neurology	2	52000
Dr. Farooq	Orthopedics	2	240000
Dr. Gupta	Orthopedics	1	130000
Dr. Hora	Neurology	1	0
Dr. Iyer	Orthopedics	1	45000
Dr. Joshi	Cardiology	1	5500

Write your SQL query below:



Q4. Find the names, wards, and ages of all patients who received at least one treatment that cost more than 50,000. Order the results alphabetically by patient name.

Expected Output:

patient_name	ward	age
Anil S.	Cardiac	70
Anita L.	Cardiac	48
Harish B.	Cardiac	63
Manoj T.	Ortho	55
Rajan H.	Ortho	52
Ramesh K.	Cardiac	45
Suresh P.	Ortho	65

Write your SQL query below:

Q5. Find all doctors whose salary is higher than the average salary of their own specialty. Show the doctor name, specialty, their salary, and the specialty average salary rounded to 2 decimal places. Order by specialty and salary descending.

Expected Output:

doctor_name	specialty	salary	specialty_avg
Dr. Dixit	Cardiology	200000	178750.00
Dr. Ahuja	Cardiology	180000	178750.00
Dr. Eapen	Neurology	210000	181666.67
Dr. Gupta	Orthopedics	195000	166000.00

Write your SQL query below:



Q6. Rank all doctors by salary within their specialty from highest to lowest. Assign the same rank to doctors with equal salaries and leave gaps in the sequence after ties. Display the doctor name, specialty, salary, and rank. Order by specialty and rank.

Expected Output:

doctor_name	specialty	salary	sal_rank
Dr. Dixit	Cardiology	200000	1
Dr. Ahuja	Cardiology	180000	2
Dr. Joshi	Cardiology	170000	3
Dr. Bose	Cardiology	165000	4
Dr. Eapen	Neurology	210000	1
Dr. Chopra	Neurology	175000	2
Dr. Hora	Neurology	160000	3
Dr. Gupta	Orthopedics	195000	1
Dr. Farooq	Orthopedics	155000	2
Dr. Iyer	Orthopedics	148000	3

Write your SQL query below:

Q7. For each patient who has at least one treatment, show every treatment row with the treatment name, date, cost, and a running cumulative total of that patient's treatment costs ordered by treatment date. Order results by patient name and treatment date.

Expected Output:

patient_name	treatment	treat_date	cost	running_cost
Anil S.	Bypass Surgery	2024-02-03	150000	150000
Anil S.	ECG	2024-02-05	2000	152000
Anita L.	Angioplasty	2024-03-12	85000	85000
Deepa V.	CT Scan	2024-03-02	9000	9000
Geeta W.	Arthroscopy	2024-04-04	45000	45000
Harish B.	Bypass Surgery	2024-05-03	150000	150000
Harish B.	ECG	2024-05-05	2000	152000
Kavita R.	EEG	2024-02-06	8000	8000
Kavita R.	MRI Brain	2024-02-10	18000	26000
Lata F.	MRI Brain	2024-05-08	18000	18000
Lata F.	EEG	2024-05-12	8000	26000
Manoj T.	Knee Replacement	2024-02-13	120000	120000
Meena C.	ECG	2024-04-16	2000	2000
Meena C.	Medication	2024-04-18	3500	5500
Rajan H.	Knee Replacement	2024-04-10	120000	120000
Ramesh K.	Angioplasty	2024-01-06	85000	85000
Ramesh K.	ECG	2024-01-08	2000	87000
Sunita M.	Echocardiogram	2024-01-09	12000	12000
Sunita M.	Medication	2024-01-12	5000	17000
Suresh P.	Hip Replacement	2024-03-07	130000	130000

Write your SQL query below:

Q8. For each treatment record, display the patient name, treatment name, treatment date, cost, the cost of that patient's previous treatment (by date), and the cost of their next treatment (by date). Show NULL where no previous or next treatment exists. Order by patient name and treatment date.

Expected Output:

patient_name	treatment	treat_date	cost	prev_cost	next_cost
Anil S.	Bypass Surgery	2024-02-03	150000	NULL	2000
Anil S.	ECG	2024-02-05	2000	150000	NULL
Anita L.	Angioplasty	2024-03-12	85000	NULL	NULL
Deepa V.	CT Scan	2024-03-02	9000	NULL	NULL
Geeta W.	Arthroscopy	2024-04-04	45000	NULL	NULL
Harish B.	Bypass Surgery	2024-05-03	150000	NULL	2000
Harish B.	ECG	2024-05-05	2000	150000	NULL
Kavita R.	EEG	2024-02-06	8000	NULL	18000
Kavita R.	MRI Brain	2024-02-10	18000	8000	NULL
Lata F.	MRI Brain	2024-05-08	18000	NULL	8000
Lata F.	EEG	2024-05-12	8000	18000	NULL
Manoj T.	Knee Replacement	2024-02-13	120000	NULL	NULL
Meena C.	ECG	2024-04-16	2000	NULL	3500
Meena C.	Medication	2024-04-18	3500	2000	NULL
Rajan H.	Knee Replacement	2024-04-10	120000	NULL	NULL
Ramesh K.	Angioplasty	2024-01-06	85000	NULL	2000
Ramesh K.	ECG	2024-01-08	2000	85000	NULL
Sunita M.	Echocardiogram	2024-01-09	12000	NULL	5000
Sunita M.	Medication	2024-01-12	5000	12000	NULL
Suresh P.	Hip Replacement	2024-03-07	130000	NULL	NULL

Write your SQL query below:

Q9. Find all patients who have no treatment records at all. Show their name, ward, and admission date. Order by patient name.

Expected Output:

patient_name	ward	admit_date
Priya N.	Neuro	2024-01-15
Vikram D.	Neuro	2024-03-18

Write your SQL query below:

Q10. Find the single most expensive treatment type in each ward. If two treatment types share the same highest cost in a ward, include both. Display the ward, treatment name, and cost. Order by ward.

Expected Output:

ward	treatment	cost
Cardiac	Bypass Surgery	150000
Neuro	MRI Brain	18000
Ortho	Hip Replacement	130000

Write your SQL query below: